MA.8.NSO.1.3

Benchmark

MA.8.NSO.1.3 **Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.**

Example: The expression $\frac{2^4}{2^7}$ is equivalent to 2^{-3} which is equivalent to $\frac{1}{8}$.

Benchmark Clarifications:
Clarification 1: Refer to the K-12 Formulas (Appendix E) for the Laws of Exponents.

Connecting Benchmarks/Horizontal Alignment

- MA.7.NSO.1.1
- MA.8.AR.1.1, MA.8.AR.1.2

Terms from the K-12 Glossary

- Exponent
- Expressions
- Integer
- Rational Numbers

Vertical Alignment

Previous Benchmarks
- MA.6.NSO.3.3

Next Benchmarks
- MA.912.NSO.1.1

Purpose and Instructional Strategies

In previous courses, students evaluated positive rational numbers and integers with natural number exponents. In grade 7 accelerated, students are introduced to the Laws of Exponents with numerical expressions with a focus on generating equivalent numerical expressions with whole-number and integer exponents and rational number bases. In Algebra 1, students will extend the Laws of Exponents to include rational exponents.

- Instruction focuses on one law at a time to allow for conceptual understanding instead of just memorizing the rules. Students should be given the opportunity to derive the properties through experience and reasoning. During instruction, include examples that show the expansion of the bases using the exponents to show the equivalence. This strategy allows for moving beyond learning a rule or procedure.
- The expectation for this benchmark includes negative integer exponents but does not include fractional exponents.

- Students should develop and engage in understanding the rules of exponents from exploration. A strategy for developing meaning for integer exponents by making use of patterns is shown below:

Patterns in Exponents	
5^5	$5 \cdot 5 \cdot 5 \cdot 5 \cdot 5$
5^4	$5 \cdot 5 \cdot 5 \cdot 5$
5^3	$5 \cdot 5 \cdot 5$
5^2	$5 \cdot 5$
5^1	5
5^0	1
5^{-1}	$\frac{1}{5}$
5^{-2}	$\frac{1}{5 \cdot 5}$
5^{-3}	$\frac{1}{5 \cdot 5 \cdot 5}$
5^{-4}	$\frac{1}{5 \cdot 5 \cdot 5 \cdot 5}$
5^{-5}	$\frac{1}{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}$

- Students should develop fluency with and without the use of a calculator when evaluating numerical expressions involving the Laws of Exponents.
- Instruction includes cases where students must work backward as well as cases where the value of a variable must be determined (*MTR.3.1*). Students should use relational thinking as well as algebraic thinking.

Common Misconceptions or Errors

- When working with negative exponents, students may not understand the connection to fractions and values in the denominator. To address this misconception, use expanded notation to show how to simplify to help support the understanding of exponents and values in the denominator of a fraction.
  - For example, $\left(\frac{5}{4}\right)^{-3}$ can be rewritten as $\left(\left(\frac{5}{4}\right)^{-1}\right)^3$ which can be rewritten as $\left(\frac{1}{\frac{5}{4}}\right)^3$ which can be rewritten as $\left(\frac{4}{5}\right)^3$ which can be rewritten as $\left(\frac{4}{5}\right)\left(\frac{4}{5}\right)\left(\frac{4}{5}\right)$ which is equivalent to $\frac{64}{125}$.

Strategies to Support Tiered Instruction

- Instruction includes teacher modeling the use expanded notation to show how to simplify. Have students practice the properties by generating equivalent expressions.
    - For example, $4^2 \times 4^{-6} = \frac{1}{4^4}$ which equals $\frac{1}{256}$ or $4 \times 4 \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{256}$. Help students to discover how 2^{-3} becomes a positive exponent in the denominator of a fraction, $\frac{1}{2^3}$.
- Instruction includes the use of a conceptual approach as opposed to memorizing the rules of the laws of integer exponents. Provide examples of the processes that lead to the rules for each law, such as the "Patterns in Exponents" table within Instructional Strategies.
- Instruction includes using expanded notation to show how to simplify to help support the understanding of exponents and values in the denominator of a fraction.
    - For example, $\left(\frac{5}{4}\right)^{-3}$ can be rewritten as $\left(\left(\frac{5}{4}\right)^{-1}\right)^3$, which can be rewritten as $\left(\frac{1}{\frac{5}{4}}\right)^3$, which can be rewritten as $\left(\frac{4}{5}\right)^3$, which can be rewritten as $\left(\frac{4}{5}\right)\left(\frac{4}{5}\right)\left(\frac{4}{5}\right)$, which is equivalent to $\frac{64}{125}$.

Instructional Tasks

Instructional Task 1 (MTR.1.1)

Create an example that will show and explain the difference between $-b$ and b^{-1}.

Instructional Task 2 (MTR.5.1)

Create a pattern using the expanded form of the base, 4, between 4^{-5} and 4^5. Explain why 4^0 is equal to 1.

Instructional Task 3 (MTR.1.1)

Aryella states that 10^0 is equivalent to 134^0. Do you agree or disagree? Explain.

Instructional Items

Instructional Item 1

What is the value of $\left(\frac{3^6}{3^{-4}}\right)^2$?

Instructional Item 2

What is the value of the expression given below?

$$\left(-\frac{2}{3}\right)^{-3}(0.8)^2$$

Instructional Item 3

Which of the following expressions are equivalent to $\frac{1}{2^6}$?

a. $2^{-5} \cdot 2^{-1}$
b. $2^{-2} \cdot 2^{-4}$
c. $2^{1} \cdot 2^{5}$
d. $2^{1} \cdot 2^{6}$
e. $2^{2} \cdot 2^{-8}$
f. $2^{2} \cdot 2^{3}$

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*